

BackEnd

Framework: Python FastAPI

Deployment Setup

Supabase:

- PostgreSQL database (managed)
- Auth (built-in JWT)
- Row-Level Security (RLS) for multi-tenanc
- Database Schema

FastAPI Backend:

- Railway, Render, or AWS Lambda
- Environment variables: DB_URL, ANTHROPIC_API_KEY, STRIPE_KEY
- CI/CD: GitHub Actions (auto-test + deploy)

Deployment:

- FrontEnd: Vercel
- BackEnd: Railway
- Database: Supabase

Need	Tool	Why
Framework	FastAPI	Fast, async, auto-docs

Auth	Supabase Auth	In Supabase
Validation	Pydantic	Type-safe request/response
Migrations	Alembic	Schema versioning
Testing	Pytest	Industry Standard
Environment	Python-dotenv	Manage secrets
Payment	Stripe-python	stripe integration
Logging	Python logging + Sentry	Error Tracking
CORS	fastapi-cors	Allow frontend requests
Rate limiting	Slowapi	Prevent abuse

Backend architecture Example

backend/

- ├── app.py # FastAPI entry point
- ├── config.py # Settings (DB, API keys, env)
- ├── requirements.txt # Dependencies
- ├── .env # Secrets (API keys, DB URL)
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- ├── api/

- | |—— routes/
 - | | |—— auth.py # Authentication endpoints
 - | | |—— restaurants.py # Restaurant CRUD
 - | | |—— inventory.py # Inventory endpoints
 - | | |—— recipes.py # Recipe endpoints
 - | | |—— agents.py # Agent execution
 - | | |—— billing.py # Stripe integration
- | |—— middleware/
 - | | |—— auth.py # JWT verification
 - | | |—— tenant.py # Multi-tenancy (isolate by restaurant)
- | |—— dependencies.py # Shared dependencies (DB, auth)
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- |—— models/
 - | |—— user.py # User schema
 - | |—— restaurant.py # Restaurant schema
 - | |—— inventory.py # Inventory schema
 - | |—— billing.py # Subscription schema
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- |—— services/
 - | |—— auth_service.py # User auth logic
 - | |—— restaurant_service.py # Restaurant operations
 - | |—— agent_service.py # Agent orchestration
 - | |—— billing_service.py # Stripe integration
- |
- |—— db/
 - | |—— connection.py # Supabase connection
 - | |—— migrations/ # Alembic migrations

| └── models.py # SQLAlchemy ORM models

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└── tests/

│ ├── test_auth.py

│ ├── test_restaurants.py

│ └── test_agents.py